

Daehyun Choi

Postdoctoral Researcher · Fluid Mechanics · Bio-inspired Propulsion · Physical AI

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“Translating biological locomotion into physical intelligence through fluid physics and machine learning”

Summary

I am a fluid mechanist and postdoctoral researcher at the Bhamla Lab, **Georgia Institute of Technology** (advisor: Prof. M. Saad Bhamla). My research decodes the fluid physics of biological locomotion — from squid propulsion to mudskipper water-hopping — and translates those principles into soft robots, underwater vehicles, and physical AI systems, using high-speed experiments, deep-learning flow diagnostics, and data-driven optimization.

Highlights

- Journals: *J. Fluid Mech.* ($\times 3$), *Phys. Fluids* (Editor’s Pick), *Sci. Rep.*, *J. R. Soc. Interface*, *Integr. Comp. Biol.*, *J. Membr. Sci.*; papers in preparation/under review (arXiv preprints available).
- Selected awards & grants (full list in Awards section): Technical leader, DARPA Young Faculty Award (PI: Prof. M. S. Bhamla, \$500k), 2024; Sejong International Postdoctoral Fellowship, NRF Korea (\$140k), 2023; Domestic Postdoctoral Scholarship, NRF Korea (\$50k), 2022.

Research Interests

- Bioinspired propulsion and aquatic locomotion
- Fluid-structure interaction and vortex dynamics
- Deep learning for flow measurement and optimization
- Multiphase flows and interfacial dynamics
- Physical AI and soft robotics

Skills

- *Experimental*: High-speed imaging, PIV, 3D-PTV, refractive-index matching, stereo-camera reconstruction, dynamometry; biological fieldwork (mudskipper tracking in natural habitats, Brunei).
- *Computational*: CNN / optical flow (PyTorch) — DeepBubbleVelocimetry [🔗★9], PWC-Net-based velocity measurement for bubble-laden flows [paper]; open-source FSI simulation (OpenFOAM + CalculiX + preCICE) [arXiv]; deep-learning Bayesian surrogate model for nozzle thrust optimization (*in preparation*); multiphysics simulation (MATLAB, Python).

Education

Seoul National University

M.S. & Ph.D. in Mechanical Engineering

Seoul, Korea

2022

Seoul National University

B.S. in Mechanical Engineering

Seoul, Korea

2015

Academic Employment

Georgia Institute of Technology

Postdoctoral Researcher

Atlanta, GA, USA

2023 – present

Ajou University

Postdoctoral Researcher

Suwon, Korea

2023

Publications

Bold denotes the presenting/corresponding author. [†] denotes equal contribution. * denotes co-first authorship.

Journal Articles

1. **Daehyun Choi**, Kai Lauren Yung, Halley Wallace, Jacob Harrison, & M. Saad Bhamla, “Interfacial dynamics on the water-hopping behavior of mudskipper,” *In preparation*.
2. **Daehyun Choi**, Paras Singh, Halley Wallace, Chandan Bose, & M. Saad Bhamla, “Deep-learning Bayesian surrogate model for thrust enhancement of soft nozzle,” *Phys. Rev. Fluids*. *Under review*.
3. **Daehyun Choi***, Paras Singh, Ian Bergerson, Minh Kim, Jieun Park, Halley J. Wallace, Kenny Zhang, Sandy Y. Hsieh, Aqua T. Asberry, Theodore A. Uyeno, William F. Gilly, Hyungmin Park, Daeshik Kang, Chandan Bose, & M. Saad Bhamla, “Squid-inspired soft superpropulsion,” *arXiv:2605.03239* (2026). *Under review*. [arXiv](#)
4. Paras Singh, **Daehyun Choi**, M. Saad Bhamla, & Chandan Bose, “Elastic wave propagation governs impulse enhancement in pulsed jets through flexible nozzles,” *arXiv:2605.17319* (2026). *Under review*. [arXiv](#)
5. Pankaj Rohilla, Johnathan Nathaniel O’Neil, Chandan Bose, Victor M. Ortega-Jimenez, **Daehyun Choi**, & M. Saad Bhamla, “Epineuston vortex recapture enhances thrust in tiny water skaters,” *Under review*. [PDF](#) [bioRxiv](#)
6. **Daehyun Choi**, Kai Yung, Ian Bergerson, Halley Wallace, Ulmar Grafe, & M. Saad Bhamla, “Three-dimensional tracking method for water-hopping mudskippers in natural habitats,” *Integr. Comp. Biol.*, 15, 2 (2025). [PDF](#) [DOI](#)
7. Pankaj Rohilla*, **Daehyun Choi***, Halley Wallace, Kai Lauren Yung, Juhi Deora, Atharva Lele, & M. Saad Bhamla (*equal contribution), “Mastering the Manu – How humans create large splashes,” *J. R. Soc. Interface*, 15, 2 (2025). [PDF](#) [DOI](#)
8. **Daehyun Choi** & Hyungmin Park, “Enhanced impulse and entrainments of the pulsed circular jet by flexible nozzle,” *J. Fluid Mech.*, 996, A6 (2024). [PDF](#) [DOI](#)
9. **Daehyun Choi***, Hyunseok Kim*, & Hyungmin Park (*equal contribution), “Bubble velocimetry using the conventional and CNN-based optical flow algorithms,” *Sci. Rep.*, 12, 11879 (2022). [PDF](#) [DOI](#) [Code](#)
10. **Daehyun Choi** & Hyungmin Park, “Flow-structure interaction of the starting jet through the flexible circular nozzle,” *J. Fluid Mech.*, 949, A39 (2022). [PDF](#) [DOI](#)
11. **Daehyun Choi**, Jungwon Byun, & Hyungmin Park, “Analysis of the liquid column atomization by annular dual-nozzle gas jet flow,” *J. Fluid Mech.*, 943, A25 (2022). [PDF](#) [DOI](#)
12. Dokeun Lee, **Daehyun Choi**, Hyungmin Park, & Sungjae Kim, “Electroconvective circulating flows by asymmetric coulombic force distribution in multiscale porous membrane,” *J. Membr. Sci.*, 636, 119286 (2021). [PDF](#) [DOI](#)
13. **Daehyun Choi** & Hyungmin Park, “Flow around in-line sphere array at moderate Reynolds number,” *Phys. Fluids (Editor’s Pick)*, 30, 097104 (2018). [PDF](#) [DOI](#)

Conferences & Seminars

1. “Hydrodynamics of water-hopping behavior of mudskippers,” 2026 Society for Integrative and Comparative Biology (SICB), Oral presentation (Symposium). *USA*, 2026.
2. “Squid-inspired flexible nozzles improve thrust in rotary propulsors using a multi-fidelity approach,” 2026 Society for Integrative and Comparative Biology (SICB), Oral presentation. *USA*, 2026.
3. “Squid funnel-inspired passively flexible nozzles for underwater propulsion,” 2026 Society for Integrative and Comparative Biology (SICB), Oral presentation. *USA*, 2026.
4. “Geometrical effects of free-end nozzles on wave reflection in self-oscillating flexible nozzles,” 78th Annual Meeting of the APS Division of Fluid Dynamics, Oral presentation. *Houston, USA*, 2025.
5. “Bio-inspired flexible nozzle design for enhanced underwater propulsion,” Fluid and Biodynamics Seminar, Max Planck Institute for Dynamics and Self-Organization, Oral presentation (Invited). *Göttingen, Germany*, 2025.

6. "Squid-inspired nozzles for enhanced thrust in rotary propulsors," 2025 *European Mechanics Society: Bio-inspired Fluid-Structure Interaction (EUROMECH)*, Oral presentation. **Vienna, Austria**, 2025.
7. "Squid-inspired nozzles for enhanced thrust in rotary propulsors," 2025 *American Physical Society March Meeting*, Oral presentation. **Anaheim, USA**, 2025.
8. "Non-invasive 3D tracking of water-hopping mudskippers in natural habitats," 2025 *Society for Integrative and Comparative Biology (SICB)*, Poster presentation. **Atlanta, USA**, 2025.
9. "Physical modelling of water-hopping mudskipper," 77th *Annual Meeting of the APS Division of Fluid Dynamics*, Poster presentation. **Salt Lake City, USA**, 2024.
10. "Water-hopping behavior of mudskipper," 2024 *IUTAM Symposium on Capillarity and Elastocapillarity in Biology*, Poster presentation. **Seoul, Korea**, 2024.
11. "Hydrodynamics of water-hopping mudskipper," 2024 *International Physics of Living Systems Annual Meeting*, Oral presentation. **Trieste, Italy**, 2024.
12. "Hydrodynamic study on the water-hopping behavior of mudskipper," 2024 *American Physical Society March Meeting*, Oral presentation. **Minneapolis, USA**, 2024.
13. "Vortex mechanism in microvelia-inspired water-walking for thrust optimization using robotic design," 2024 *Society for Integrative and Comparative Biology (SICB)*, Poster presentation. **Seattle, USA**, 2024.
14. "Experimental study on the starting jet through the flexible circular nozzle," 2022 *World Congress on Advances in Civil, Environmental and Materials Research (ACEM)*, Oral presentation. **Seoul, Korea**, 2022.
15. "On the optimal flexibility condition of an impulsive starting jet nozzle for thrust generation," 73rd *Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. **Online**, 2020.
16. "Experimental study on the effect of nozzle flexibility on the evolution of a starting liquid jet," 72nd *Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. **Seattle, USA**, 2019.
17. "Study on dual-nozzle jet flows for the application of liquid atomization," 71st *Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. **Atlanta, USA**, 2018.
18. "Experimental study on the flow around in-line array of spheres," 69th *Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. **Portland, USA**, 2016.

Research Project Experience

Squid-Inspired Nozzles for Enhanced Efficiency and Thrust in Rotary Propulsors Supported by Defense Advanced Research Projects Agency	Georgia Institute of Technology 2024 – present
Hydrodynamics of Water-Hopping Mudskipper and Biomimetic Robot Development Supported by National Research Foundation of Korea – Sejong International Postdoctoral Fellowship	Georgia Institute of Technology 2023 – present
Development of Cephalopod-Inspired Jet-Propelled Soft Underwater Vehicle Supported by National Research Foundation of Korea – Domestic Postdoctoral Fellowship	Ajou University 2022 – 2023
Physical-Based Model for Film Thickness and Spray Characteristics Depending on Operation Conditions Supported by Daewoo Shipbuilding & Marine Engineering (DSME)	Seoul National University 2021 – 2022
Apparatus for Determining Removal of Foley Catheter Supported by Yonsei Severance Hospital	Seoul National University 2020 – 2022

Measurement of Bubble and Particle in Liquid
Supported by Samsung Electro-Mechanics

Seoul National University
2018 – 2019

Analysis of Molten Metal Atomization Using Multi-physics Simulation
Supported by Samsung Electronics

Seoul National University
2018 – 2018

Effects of Superhydrophobic Surface on the Wake of Circular Cylinder
Supported by Samsung Advanced Institute of Technology (SAIT)

Seoul National University
2016 – 2016

Awards & Grants

- 2025 **Emerging Scholar Best Paper Prize (Top 10)**, *Journal of Fluid Mechanics*, Cambridge University Press
- 2024 **Technical Leader – DARPA Young Faculty Award (\$500k)**, Defense Advanced Research Projects Agency
- 2024 **Robert M. Nerem International Travel Award (\$3k)**, Georgia Institute of Technology
- 2024 **Young Researcher Travel Award (\$1k)**, IUTAM Symposium on Capillarity and Elastocapillarity in Biology
- 2023 **Sejong International Postdoctoral Fellowship (\$140k)**, National Research Foundation of Korea
- 2022 **Best Thesis Award**, Dept. of Mechanical Engineering, Seoul National University
- 2022 **Domestic Postdoctoral Scholarship (\$50k)**, National Research Foundation of Korea
- 2016 **BK21+ Doctoral Scholarship (\$10k)**, Ministry of Education, Korea
- 2015 **Presidential Scholarship for Undergraduates (\$30k)**, Korea
- 2014 **1st Prize – EDISON Challenge (CFD contest, \$1k)**, Ministry of Science, ICT and Future Planning, Korea

Mentoring & Teaching

Graduate (3) and undergraduate (5) students; 1 high school student
Research Mentor

Georgia Institute of
Technology
2024 – present

3 undergraduate students
Undergraduate Thesis Advisor

Seoul National University
2016 – 2021

Mechanical Engineering Laboratory: Flow Measurements
Teaching Assistant

Seoul National University
2016 – 2017

Patents (Lead Inventor)

Compliant Nozzle System and Method for Rotary Propulsion
US Patent Application, No. 19/568,682

Georgia Tech Research
Corporation
2026

Compliant Nozzle System and Method for Fluid Jet Impulse
US Patent Application, No. 19/568,695

Georgia Tech Research
Corporation
2026

System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with flow and hydraulic pressure KR Patent, 10-2021-0149171	Korea 2021
System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with hydraulic pressure and UASIF KR Patent, 10-2021-0149176	Korea 2021
System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with UASIF KR Patent, 10-2021-0149175	Korea 2021
Apparatus for determining removal of Foley catheter PCT Patent, PCT/KR2020/018848	International 2020
System for reducing ureteral insertion pressure by artificially triggering hydronephrosis based on measurement of pressure associated with the insertion of the ureteral access sheath for RIRS KR Granted Patent, 10-2020-0064729	Korea 2020
System for reducing ureteral entry pressure by generating vibrations in the guide wire for the insertion of the ureteral access sheath for RIRS KR Granted Patent, 10-2020-0064731	Korea 2020
Apparatus for determining removal of Foley catheter KR Granted Patent, 10-2020-0000288	Korea 2020